Unit 4 – Network Technology

4.5 Devices

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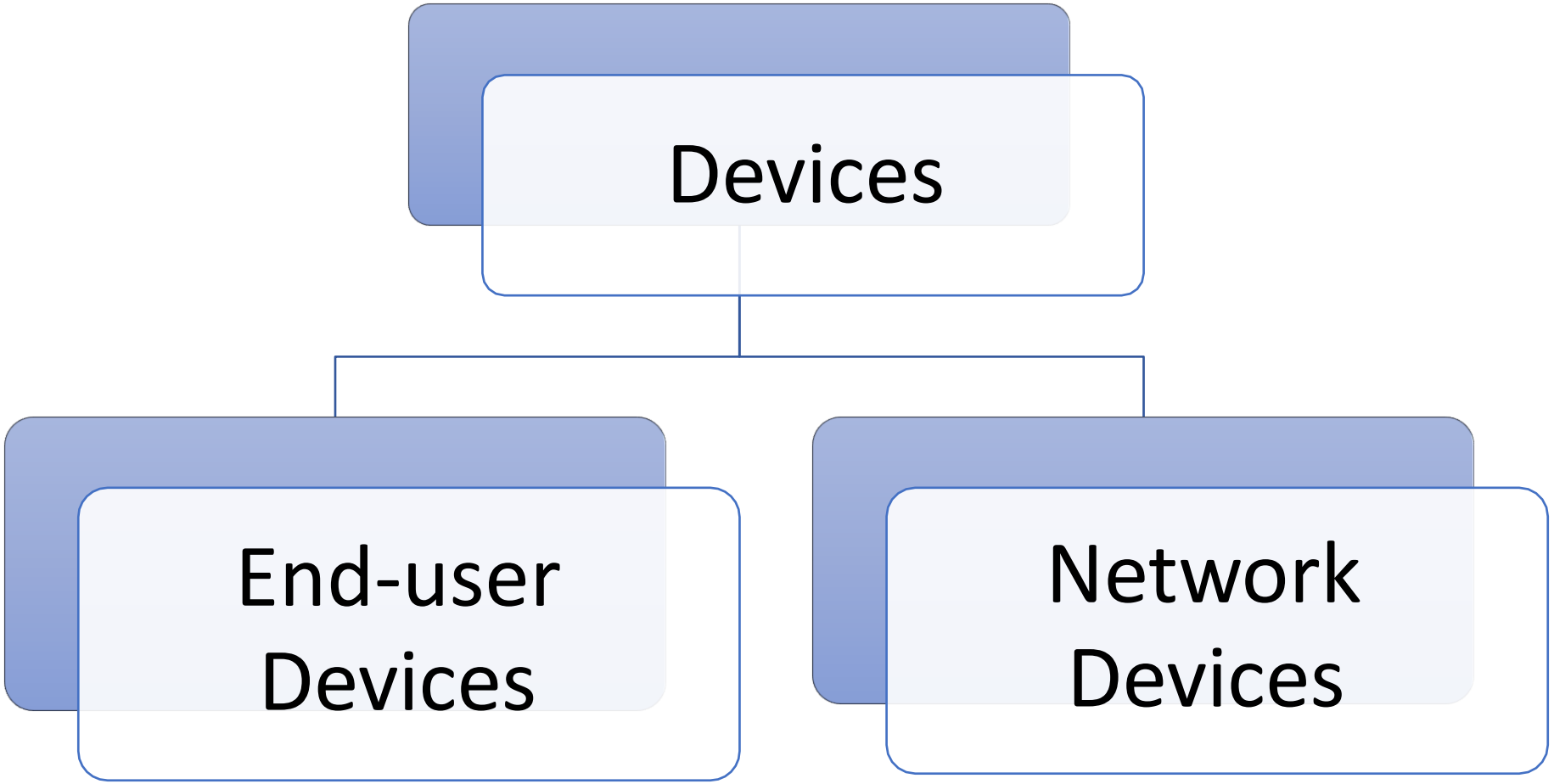

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End-user Devices

- Devices which provide services directly to the user. Example: Computers, printers, scanners.
- These devices allows users to share, create and obtain information

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Network Devices

- Devices that connect the end –user devices together to allow them to communicate.

Example: Repeaters

Bridges

Network Interface Card (NIC)

Switches

Routers

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Repeaters

- A repeater is a network device that retransmits a received signal with more power and to an extended geographical or topological network boundary than what would be capable with the original signal.

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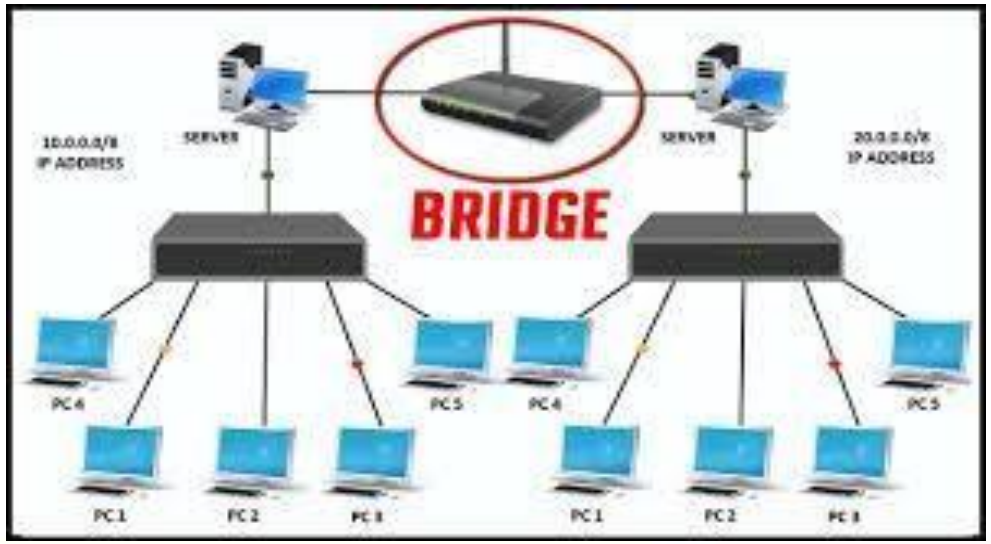
Bridges

- A **network bridge** is a device that can create a single network from different and multiple network segments.
- In other words, it can connect the two or more networks. The function used here is called **Network Bridging**.



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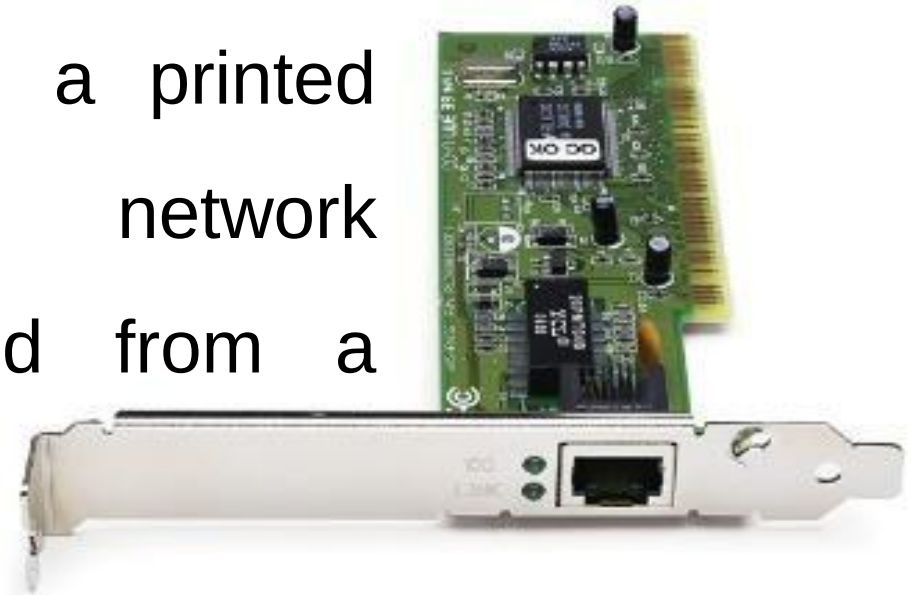
Network Interface Cards (NIC)

- A **network interface card (NIC)** is a printed circuit board that provides network communication capabilities to and from a personal computer



- Each individual NIC carries a Unique Code, calls Media Access Control (MAC) Address.

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Switches

- A switch is a device that provides a central connection point for cables from workstations, servers, and peripherals. In a star topology, twisted pair wire is run from each workstation to a central switch/hub.

- Switches manage the flow of data across a network by transmitting a received network packet only to the one or more devices for which the packet is intended.

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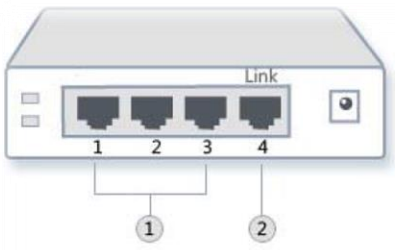
Hubs

- A hub interconnects two or more workstations into a local area network.
- When a workstation transmits to a hub, the hub immediately resends the data frame to all connecting links.
- Hubs expand one Ethernet connection into many. For example, a four-port hub connects up to four machines.



1 Ports for computers

2 Ports for connecting to a router, another hub or a switch (typically labeled "Link" / "Uplink" / "Out")

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Routers

- Routers can regenerate signals, concentrate multiple connections and manage data transfers.
- They can also connect to a WAN, which allows them to connect LANs that are separated by great distances.
- None of the other devices can provide this type of connection.

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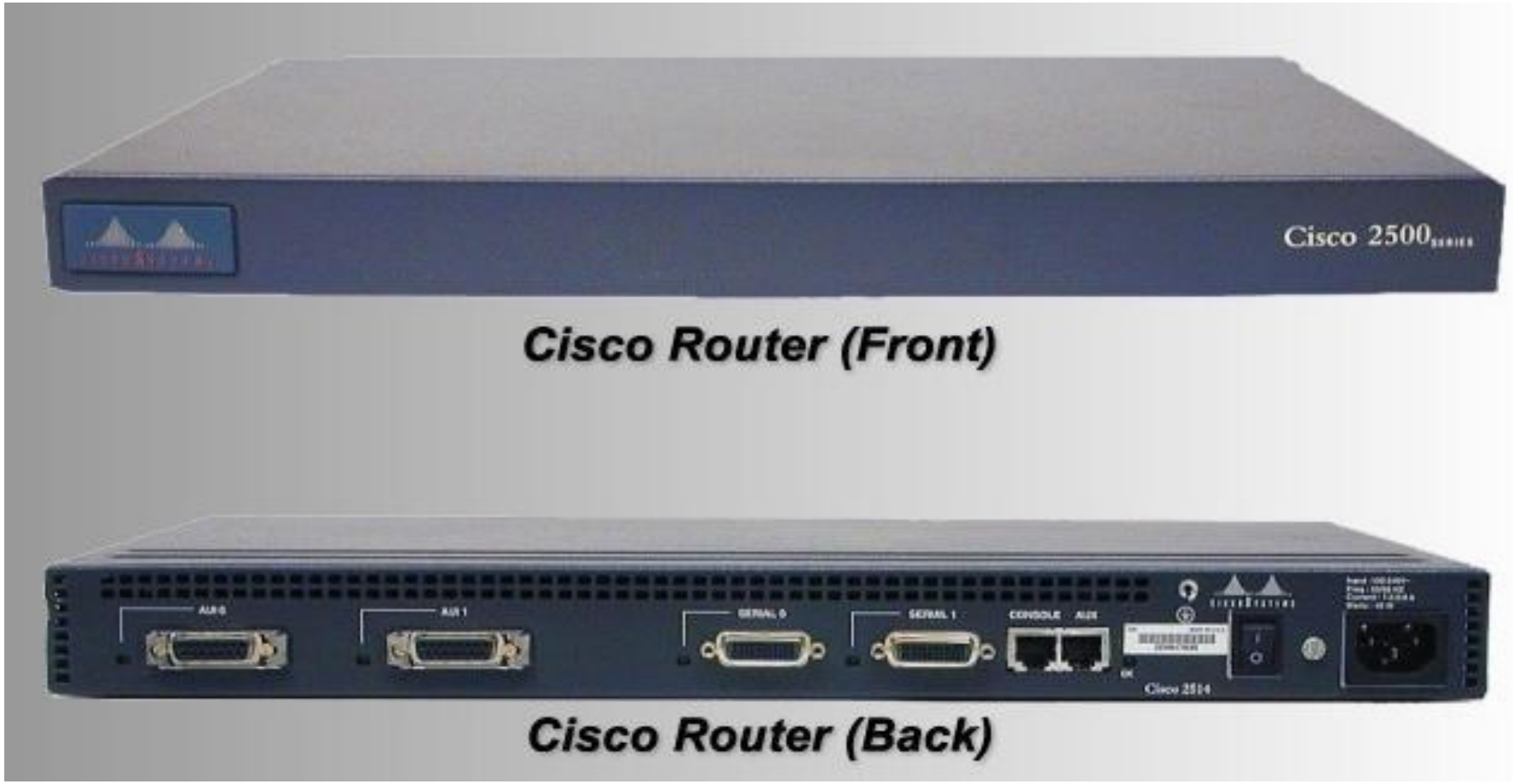
Routers



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Wireless Network Communications

- Wireless communications is a type of data communication that is performed and delivered wirelessly.
- This is a broad term that incorporates all procedures and forms of connecting and communicating between two or more devices using a wireless signal through wireless communication technologies and devices.

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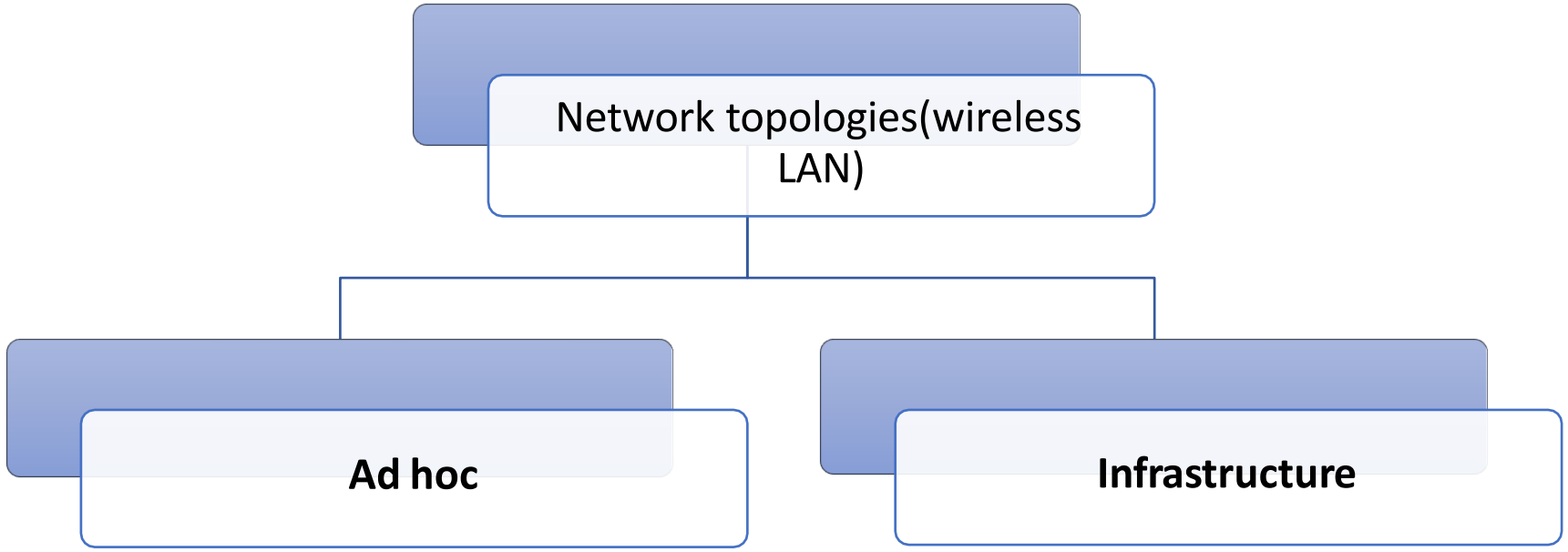
Wireless Network Communications

- There are two types of network topologies supported by wireless LANs.



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Wireless Network Communications

- Most Wi-Fi networks function in **infrastructure** mode. Devices on the network all communicate through a single access point, which is generally the wireless router.

- An **Ad-hoc** network allows each device to communicate directly with each other. Ad-hoc networks don't require a centralized access point.

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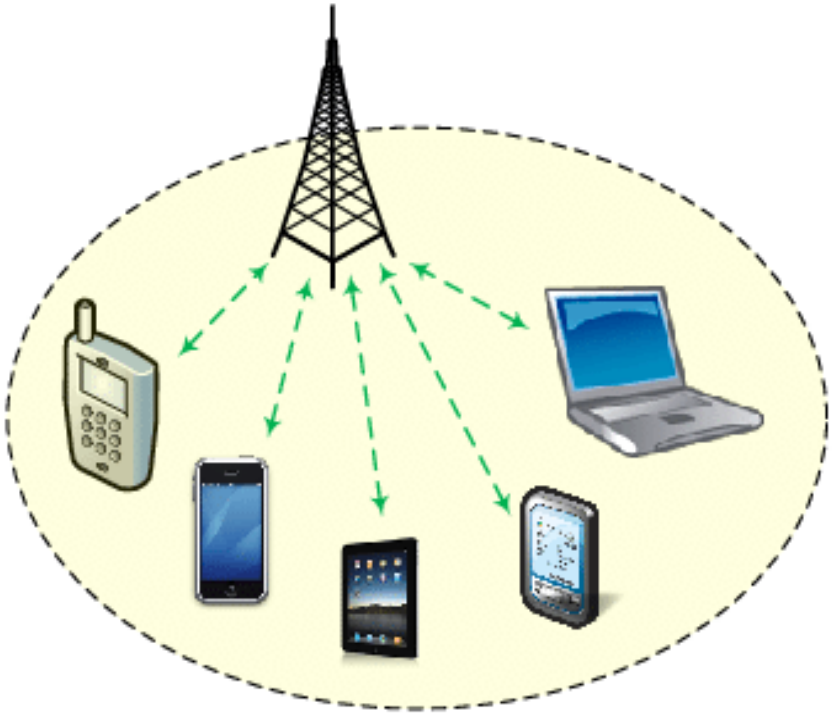
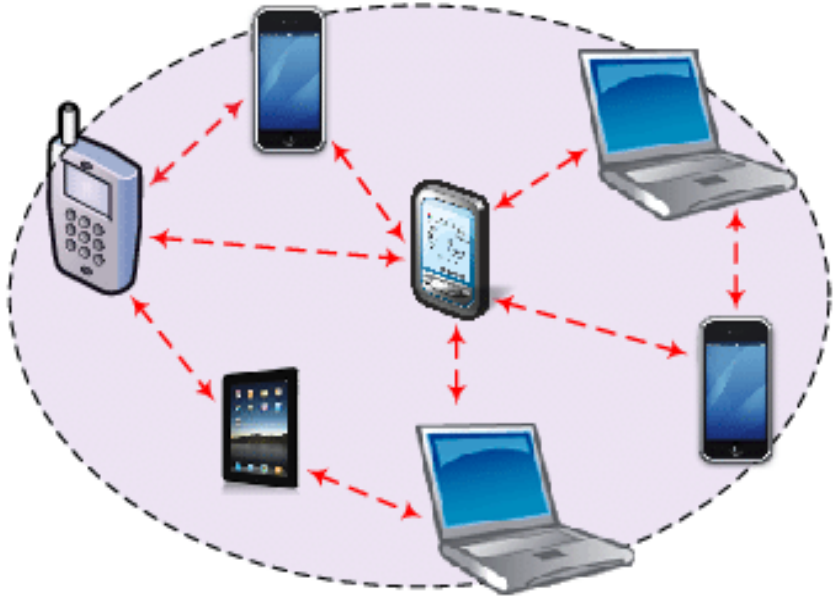
Ad hoc network vs Infrastructure network



Infrastructure-based wireless networks



Wireless ad hoc networks

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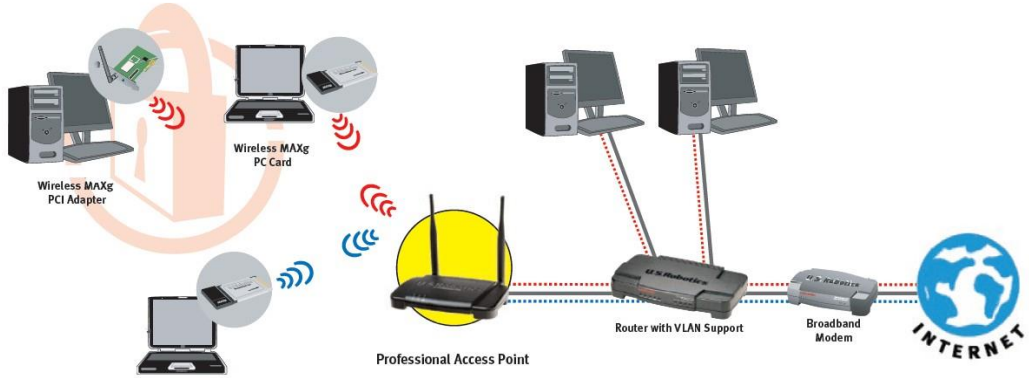
Access Point

- An access point can be considered to represent a two- port bridge. However, instead of each port begin connected to wired LAN, one port is connected to an antenna.



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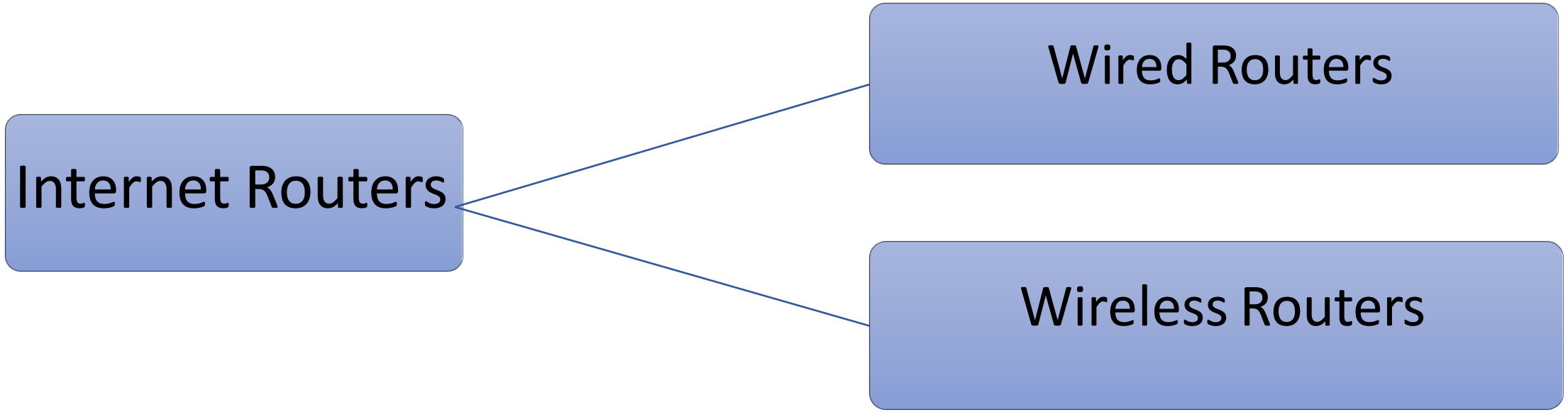
Internet Routers



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Wireless Routers

- A wireless router is an electronic device that works as a router(It sends data from the internet cable to device) and as a wireless access points so this data can be shared through radio signals instead of another cable.
- The Wi-Fi router takes the data from telephone line and converts it in to radio signals, which are then picked up by devices with Wi-Fi capability such as laptops, tablets and smartphones.

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Wireless Routers

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Wi-Fi Data Security

- Securing our Wi-Fi connections is an important element of securing your personal data. A Wi-Fi network using WPA2 provides both **security** and **privacy** for communications as they travel across your network.

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Securing a new network

- Change the network name(SSID) from the default name.
- Change the administrative credentials(username and password) that control the configuration settings of your Access Point/Router/Gateway.
- Enable WPA2-Personal(aka WPA2-PSK) with AES encryption.

- Creates network passphrase that meets recommended guidelines.
- Enable WPA2 security features on your client device and enter the passphrase for your network.

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Lesson Summery

- Network peripherals

  Repeate
  rs
  Bridges
  Network Interface Cards
  (NIC) Switch
  Routers
  Hubs

- Wireless network communication
- Wi-Fi Data Security
- Securing a new network

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